

# Project Workflow & Work Breakdown Structure (WBS)

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## EMS Readiness Optimization for Manhattan

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**Date:** March 12, 2026

**Version:** 1.0.0

**Project:** EMS Readiness Optimization — Manhattan, NYC

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## 1. Overview

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This document provides the complete Work Breakdown Structure (WBS) for the EMS Readiness Optimization project. The project follows a seven-phase methodology, progressing from problem definition through data engineering, optimization, simulation, experimentation, and final reporting. Each phase is documented with its objectives, step-by-step procedures, input/output files, scripts used, and dependencies.

**Project Scope:** Evaluate strategic EMS ambulance staging policies across 48 FDNY firehouses in Manhattan using discrete-event simulation, comparing uniform (P0), demand-proportional (P1), and MIP-optimized (P2) allocation strategies.

**Total Deliverables:** 14 Python modules, 16 scripts, 9 notebooks, 44 figures, 24 tables, 25+ documentation files.

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## 2. Phase 1 — Problem Definition & Project Setup

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### Objective

Define the decision problem, research questions, scope, and project infrastructure.

## Step-by-Step Procedure

| Step | Action                                                                                                                                 | Output               |
|------|----------------------------------------------------------------------------------------------------------------------------------------|----------------------|
| 1.1  | Define the operational decision: optimal staging of K ambulances across 48 fire-houses                                                 | Problem statement    |
| 1.2  | Formulate 5 research questions (RQ1–RQ5) covering spatial demand, optimal allocation, policy comparison, sensitivity, and fleet sizing | Research questions   |
| 1.3  | Select primary (mean response time) and secondary (P90 RT, 8-min coverage, utilization) performance measures                           | KPI definitions      |
| 1.4  | Define system boundary: Manhattan, MVC incidents, static allocation, Haversine travel proxy                                            | Scope statement      |
| 1.5  | Document assumptions, exclusions, and limitations                                                                                      | Assumptions log      |
| 1.6  | Set up repository structure, Python package, and configuration files                                                                   | Project skeleton     |
| 1.7  | Create project charter and decisions log                                                                                               | Governance documents |

## Input Files

- None (project inception)

## Output Files / Deliverables

| File                          | Description                                               |
|-------------------------------|-----------------------------------------------------------|
| docs/project_charter.md       | Project charter with objectives, scope, and timeline      |
| docs/assumptions_log.md       | Documented assumptions (12 items) with rationale and risk |
| docs/decisions_log.md         | Key decisions (DEC-001 through DEC-005)                   |
| docs/blocker_log.md           | Blocker tracking template                                 |
| docs/source_manifest.md       | Data source inventory                                     |
| requirements.txt              | Python dependency specification                           |
| Makefile                      | Build automation targets                                  |
| README.md                     | Project overview and setup instructions                   |
| .gitignore                    | Git ignore rules for large data and generated files       |
| src/ems_readiness/__init__.py | Package initialization                                    |

## Scripts / Notebooks Used

- Manual creation (no automated scripts in this phase)

## Dependencies

- None (starting phase)

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## 3. Phase 2 — Data Strategy & Engineering

### Objective

Acquire, clean, and transform raw data into analysis-ready datasets; build spatial and temporal demand models; construct the firehouse-to-precinct distance matrix.

## Step-by-Step Procedure

| Step | Action                                                                                                                            | Output                             |
|------|-----------------------------------------------------------------------------------------------------------------------------------|------------------------------------|
| 2.1  | Download raw crash records (2.24M rows), firehouse listing, precinct boundaries, and geographic boundary files from NYC Open Data | Raw data in <code>data/raw/</code> |
| 2.2  | Audit raw data quality: check boundaries, firehouse coordinates, precinct geometries, crash records                               | Audit reports                      |
| 2.3  | Spatial filtering: retain 628,811 Manhattan crashes with valid coordinates                                                        | Filtered crash dataset             |
| 2.4  | Temporal extraction: derive hour, day-of-week, month from crash timestamps                                                        | Enriched crash dataset             |
| 2.5  | Filter 48 Manhattan firehouses from 219 city-wide                                                                                 | Manhattan firehouse list           |
| 2.6  | Match 30 Manhattan precincts and compute centroids                                                                                | Precinct centroids                 |
| 2.7  | Build 48×30 Haversine distance matrix (firehouses × precincts)                                                                    | Distance matrix CSV                |
| 2.8  | Calibrate NHPP demand model: compute hourly factors, DOW factors, precinct proportions                                            | Lambda tables                      |
| 2.9  | Build travel-time proxy model (Haversine / 20 mph with TOD factors)                                                               | Travel time module                 |
| 2.10 | Build service-time distribution model (LogNormal, mean=25 min, std=10 min)                                                        | Service time module                |
| 2.11 | Perform exploratory data analysis (EDA): spatial, temporal, and statistical visualization                                         | EDA figures and notebook           |
| 2.12 |                                                                                                                                   | Model spec documents               |

| Step | Action                                                              | Output |
|------|---------------------------------------------------------------------|--------|
|      | Document demand model specification and service model specification |        |

## Input Files

| File                                                    | Source                                |
|---------------------------------------------------------|---------------------------------------|
| data/raw/Motor_Vehicle_Collisions_-Crashes_20260223.csv | NYC Open Data (536 MB, 2.24M records) |
| data/raw/FDNY_Firehouse_Listing_20260223.csv            | NYC Open Data (219 firehouses)        |
| data/raw/Police_Precincts_20260223.csv                  | NYC Open Data (78 precincts)          |
| data/raw/manhattan_boundary.geojson                     | NYC Open Data                         |
| data/raw/cbd_boundary.geojson                           | MTA Congestion Relief Zone            |
| data/raw/nyc_borough_boundaries.geojson                 | NYC Open Data                         |
| data/raw/*_Data_Dictionary.xlsx                         | NYC Open Data (3 data dictionaries)   |

## **Output Files / Deliverables**

| File                                                       | Description                                          |
|------------------------------------------------------------|------------------------------------------------------|
| data/processed/dis-<br>tance_matrix_firehouse_precinct.csv | 48×30 Haversine distance matrix (miles)              |
| data/processed/demand_lambda_hourly.csv                    | 24 hourly intensity factors for NHPP                 |
| data/processed/demand_lambda_dow.csv                       | 7 day-of-week intensity factors                      |
| data/processed/demand_lambda_precinct.csv                  | 30 precinct demand proportions                       |
| src/ems_readiness/utils/distance.py                        | Haversine distance and matrix builder                |
| src/ems_readiness/service/travel_time.py                   | Travel-time proxy with TOD factors                   |
| src/ems_readiness/service/service_time.py                  | LogNormal service-time sampler                       |
| src/ems_readiness/demand/ar-<br>rival_generator.py         | NHPP arrival generator (Lewis-Shedler thin-<br>ning) |
| configs/demand.yaml                                        | Demand model configuration                           |
| configs/service.yaml                                       | Service model configuration                          |
| docs/demand_model_spec.md                                  | Demand model specification document                  |
| docs/service_model_spec.md                                 | Service & travel proxy specification                 |
| docs/eda_and_data_split_summary.md                         | EDA summary and data split documentation             |
| results/figures/fig_crash_heatmap.png                      | Crash heatmap                                        |
| results/figures/fig_hourly_demand.png                      | Hourly demand profile                                |
| results/figures/fig_daily_demand.png                       | Daily demand profile                                 |
| results/figures/fig_firehouses_map.png                     | Manhattan firehouses map                             |
| results/figures/fig_precinct_demand.png                    | Precinct demand bar chart                            |
| results/figures/fig_precinct_density.png                   | Precinct density choropleth                          |
| results/figures/fig_temporal_trends.png                    | Long-term temporal trends                            |
| results/figures/fig_hourly_rates.png                       | NHPP hourly rate factors                             |
| results/figures/fig_demand_model_fit.png                   | Demand model fit diagnostics                         |
| results/figures/dis-<br>tance_matrix_heatmap.png           | Distance matrix heatmap                              |



| File                                          | Description                |
|-----------------------------------------------|----------------------------|
| results/figures/travel_time_by_tod.png        | Travel time by time-of-day |
| results/figures/tod_speed_factors.png         | TOD speed factor profile   |
| results/figures/service_time_distribution.png | Service time distribution  |
| results/figures/nhpp_arrivals_demo.png        | NHPP arrival demo          |

## Scripts / Notebooks Used

| Script / Notebook                       | Purpose                                              |
|-----------------------------------------|------------------------------------------------------|
| scripts/data_audit.py                   | Comprehensive raw data quality audit                 |
| scripts/audit_step1_boundaries.py       | Audit geographic boundaries                          |
| scripts/audit_step2_firehouses.py       | Audit firehouse data                                 |
| scripts/audit_step3_precincts.py        | Audit precinct data                                  |
| scripts/audit_step4_crashes.py          | Audit crash records                                  |
| scripts/demand_modeling.py              | Calibrate NHPP demand model and export lambda tables |
| notebooks/02_eda_spatiotemporal.ipynb   | Exploratory data analysis with visualizations        |
| notebooks/03_input_modeling.ipynb       | Input modeling and demand estimation                 |
| notebooks/04_service_travel_proxy.ipynb | Service/travel model demonstration and validation    |

## Dependencies

- **Depends on:** Phase 1 (project structure, configuration framework)

## 4. Phase 3 — Optimization & Policy Design

### Objective

Formulate and solve MIP allocation models to generate candidate staging policies (P0, P1, P2, P2-alt, P2-cov) for simulation evaluation.

## Step-by-Step Procedure

| Step | Action                                                                                                                                        | Output               |
|------|-----------------------------------------------------------------------------------------------------------------------------------------------|----------------------|
| 3.1  | Define P0 (uniform) baseline allocation: $K / 48$ with round-robin remainder                                                                  | Baseline policy      |
| 3.2  | Implement P1 (demand-proportional) heuristic: allocate proportional to nearest-fire-house demand credit                                       | Heuristic policy     |
| 3.3  | Formulate demand-weighted MIP (P2): minimize $\sum d_j \cdot t_{ij} \cdot y_{ij}$ subject to unit total, capacity, and assignment constraints | P2 formulation       |
| 3.4  | Formulate P-median model (P2-alt): select $K$ firehouses to minimize weighted distance                                                        | P2-alt formulation   |
| 3.5  | Formulate maximal coverage model (P2-cov): maximize demand within 8-minute threshold                                                          | P2-cov formulation   |
| 3.6  | Implement <code>EMSAAllocator</code> class as high-level solver interface                                                                     | Solver module        |
| 3.7  | Solve all models for $K \in \{20, 30, 40, 48\}$ and compare results                                                                           | Comparison tables    |
| 3.8  | Perform sensitivity analysis of objective values vs. fleet size                                                                               | Sensitivity figures  |
| 3.9  | Validate all policies for feasibility (unit totals, capacity caps)                                                                            | Feasibility checks   |
| 3.10 | Document mathematical formulations                                                                                                            | Formulation document |

## Input Files

| File                                                   | Source                     |
|--------------------------------------------------------|----------------------------|
| data/processed/dis-tance_matrix_firehouse_precinct.csv | Phase 2 output             |
| data/processed/demand_lambda_precinct.csv              | Phase 2 output             |
| configs/optimization.yaml                              | Configuration              |
| configs/service.yaml                                   | Travel speed configuration |

## Output Files / Deliverables

| File                                          | Description                                                    |
|-----------------------------------------------|----------------------------------------------------------------|
| src/ems_readiness/optimization/models.py      | MIP formulations (demand-weighted, p-median, maximal coverage) |
| src/ems_readiness/optimization/policies.py    | Baseline policies (uniform, demand-proportional)               |
| src/ems_readiness/optimization/allocator.py   | High-level EMSAllocator solver class                           |
| configs/optimization.yaml                     | Optimization configuration                                     |
| docs/optimization_formulation.md              | Mathematical specification of all models                       |
| docs/optimization_results.md                  | Optimization results and comparison                            |
| results/tables/optimization_comparison.csv    | Model comparison table                                         |
| results/figures/opt_allocation_comparison.png | Allocation comparison visualization                            |
| results/figures/opt_inputs.png                | Optimization inputs visualization                              |
| results/figures/opt_sensitivity.png           | Sensitivity analysis figure                                    |

## Scripts / Notebooks Used

| Script / Notebook                                   | Purpose                                             |
|-----------------------------------------------------|-----------------------------------------------------|
| <code>scripts/run_optimization_comparison.py</code> | Run all optimization models and generate comparison |
| <code>notebooks/05_optimization.ipynb</code>        | Interactive optimization demo and analysis          |
| <code>notebooks/05_optimization.py</code>           | Source script for optimization notebook             |

## Dependencies

- **Depends on:** Phase 2 (distance matrix, demand data, travel time model)

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## 5. Phase 4 — Simulation Design & Implementation

### Objective

Design the conceptual DES model and implement a modular simulation engine in Python using SimPy.

## **Step-by-Step Procedure**

| Step | Action                                                                                                        | Output                 |
|------|---------------------------------------------------------------------------------------------------------------|------------------------|
| 4.1  | Define entities (Incident), resources (EMSUnit), queues (FIFO), and state variables                           | Conceptual model       |
| 4.2  | Specify 5 event types: Arrival, Dispatch, Service Start, Service Completion, End of Simulation                | Event logic            |
| 4.3  | Design nearest-available dispatch logic with Haversine travel-time estimation                                 | Dispatch specification |
| 4.4  | Define performance measures: mean RT, P90 RT, 8-min coverage, utilization, queue metrics                      | KPI specification      |
| 4.5  | Document random phenomena: NHPP arrivals (Stream 1), precinct assignment (Stream 2), service times (Stream 3) | RNG strategy           |
| 4.6  | Create event flow diagram and single-incident timeline                                                        | Flowcharts             |
| 4.7  | Implement Incident data-class with full lifecycle tracking                                                    | entities.py            |
| 4.8  | Implement EMSUnit, UnitPool, and UnitStatus for resource management                                           | resources.py           |
| 4.9  | Implement NearestAvailableDispatcher with cached travel-time matrices                                         | dispatcher.py          |
| 4.10 | Implement MetricsCollector for KPI tracking                                                                   | metrics.py             |
| 4.11 | Implement EMSSimulation main engine with SimPy process-interaction                                            | engine.py              |
| 4.12 |                                                                                                               | runner.py              |

| Step | Action                                                            | Output        |
|------|-------------------------------------------------------------------|---------------|
|      | Implement BatchRunner for replicated experiments with CRN support |               |
| 4.13 | Write comprehensive unit tests (39 tests across 4 modules)        | Test suite    |
| 4.14 | Document conceptual model and code architecture                   | Documentation |

## Input Files

| File                                                   | Source                           |
|--------------------------------------------------------|----------------------------------|
| data/processed/dis-tance_matrix_firehouse_precinct.csv | Phase 2 output                   |
| data/processed/demand_lambda_*.csv                     | Phase 2 output (3 lambda tables) |
| configs/simulation.yaml                                | Simulation configuration         |
| configs/demand.yaml                                    | Demand model parameters          |
| configs/service.yaml                                   | Service model parameters         |

## Output Files / Deliverables

| File                                                    | Description                                                            |
|---------------------------------------------------------|------------------------------------------------------------------------|
| <code>src/ems_readiness/simulation/__init__.py</code>   | Simulation package initialization                                      |
| <code>src/ems_readiness/simulation/entities.py</code>   | Incident dataclass                                                     |
| <code>src/ems_readiness/simulation/resources.py</code>  | <code>EMSUnit</code> , <code>UnitPool</code> , <code>UnitStatus</code> |
| <code>src/ems_readiness/simulation/dispatcher.py</code> | <code>NearestAvailableDispatcher</code>                                |
| <code>src/ems_readiness/simulation/metrics.py</code>    | <code>MetricsCollector</code>                                          |
| <code>src/ems_readiness/simulation/engine.py</code>     | <code>EMSSimulation</code> main DES engine                             |
| <code>src/ems_readiness/simulation/runner.py</code>     | <code>BatchRunner</code> for experiment execution                      |
| <code>configs/simulation.yaml</code>                    | DES engine configuration                                               |
| <code>docs/conceptual_model.md</code>                   | Complete conceptual model specification                                |
| <code>docs/code_documentation.md</code>                 | Code architecture and module documentation                             |
| <code>tests/</code>                                     | 39 unit tests across 4 test modules                                    |

## Scripts / Notebooks Used

| Script / Notebook                                | Purpose                               |
|--------------------------------------------------|---------------------------------------|
| <code>notebooks/06_simulation_debug.ipynb</code> | Simulation debugging and validation   |
| <code>notebooks/06_simulation_debug.py</code>    | Source script for simulation notebook |

## Dependencies

- **Depends on:** Phase 2 (demand model, distance matrix), Phase 3 (allocation policies)

## 6. Phase 5 — Verification & Validation

### Objective

Verify that the simulation code correctly implements the conceptual model and validate that the model produces credible results.



## Step-by-Step Procedure

| Step | Action                                                                                                                                 | Output                 |
|------|----------------------------------------------------------------------------------------------------------------------------------------|------------------------|
| 5.1  | <b>Verification Test 1 — Toy Example:</b> Run a small deterministic scenario with known analytical solution; hand-trace event timeline | Toy timeline figure    |
| 5.2  | <b>Verification Test 2 — Zero Demand:</b> Set $\lambda_0 = 0$ and verify no incidents are generated                                    | Zero-demand check      |
| 5.3  | <b>Verification Test 3 — Single-Unit Saturation:</b> Run with $K=1$ under high demand; verify queueing occurs correctly                | Saturation test        |
| 5.4  | <b>Verification Test 4 — Extreme Demand:</b> Stress test under $10\times$ demand to verify queue accumulation and system stability     | Extreme demand test    |
| 5.5  | <b>Validation Pilot 1 — Directional:</b> Compare $P_0$ vs. $P_2$ and confirm $P_2$ dominates on mean RT                                | Directional validation |
| 5.6  | <b>Validation Pilot 2 — Fleet Sensitivity:</b> Verify response time decreases monotonically with increasing $K$                        | Monotonicity check     |
| 5.7  | <b>Validation Pilot 3 — Demand Sensitivity:</b> Verify response time increases with demand intensity                                   | Demand response check  |
| 5.8  | Run all 39 unit tests and confirm 100% pass rate                                                                                       | Test results           |
| 5.9  | Document all V&V findings, corrections, and acceptance criteria                                                                        | V&V log                |

## Input Files

| File                                 | Source                   |
|--------------------------------------|--------------------------|
| All Phase 2 processed data           | Phase 2 output           |
| All Phase 3 allocation policies      | Phase 3 output           |
| <code>configs/simulation.yaml</code> | Simulation configuration |

## Output Files / Deliverables

| File                                                           | Description                        |
|----------------------------------------------------------------|------------------------------------|
| <code>docs/verification_log.md</code>                          | Complete V&V results documentation |
| <code>results/figures/verification_toy_timeline.png</code>     | Toy example event timeline         |
| <code>results/figures/validation_p0_vs_p2.png</code>           | Validation: P0 vs. P2 comparison   |
| <code>results/figures/validation_sensitivity_K.png</code>      | Validation: Fleet sensitivity      |
| <code>results/figures/validation_sensitivity_demand.png</code> | Validation: Demand sensitivity     |

## Scripts / Notebooks Used

| Script / Notebook                             | Purpose                                    |
|-----------------------------------------------|--------------------------------------------|
| <code>scripts/run_verification.py</code>      | Execute all 4 verification tests (seed=42) |
| <code>scripts/run_validation_pilots.py</code> | Execute all 3 validation pilots            |

## Dependencies

- **Depends on:** Phase 4 (simulation engine), Phase 3 (allocation policies), Phase 2 (data)

# 7. Phase 6 — Experimentation & Analysis

## Objective

Execute production simulation experiments, perform statistical analysis, and generate publication-quality results.

## Step-by-Step Procedure

| Step | Action                                                                                    | Output                |
|------|-------------------------------------------------------------------------------------------|-----------------------|
| 6.1  | Freeze experimental design: 4 main experiments + CBD robustness                           | Experiment matrix     |
| 6.2  | <b>Exp 1 — Policy Comparison:</b> 3 policies × 30 replications = 90 runs at K=20          | Exp1 results          |
| 6.3  | <b>Exp 2 — Fleet Sensitivity:</b> 3 policies × 6 fleet sizes × 30 reps = 540 runs         | Exp2 results          |
| 6.4  | <b>Exp 3 — Demand Sensitivity:</b> 3 policies × 6 demand multipliers × 30 reps = 540 runs | Exp3 results          |
| 6.5  | <b>Exp 4 — Service Robustness:</b> 3 policies × 3 service times × 30 reps = 270 runs      | Exp4 results          |
| 6.6  | Execute all 1,440 production runs                                                         | Raw simulation output |
| 6.7  | <b>CBD Experiment:</b> 11 CBD scenarios × 30 replications = 330 additional runs           | CBD results           |
| 6.8  | Compute descriptive statistics, ANOVA, Tukey HSD, effect sizes, and CIs                   | Statistical tables    |
| 6.9  | Analyze queue metrics across all 1,770 runs                                               | Queue analysis        |
| 6.10 | Analyze seasonal variation patterns                                                       | Seasonal analysis     |
| 6.11 | Generate publication-quality figures (5 main figures)                                     | Publication figures   |
| 6.12 | Generate project summary dashboard                                                        | Dashboard figure      |
| 6.13 | Document experimental design and output analysis                                          | Design document       |

## Input Files

| File                                      | Source                       |
|-------------------------------------------|------------------------------|
| All processed data from Phase 2           | Phase 2 output               |
| Allocation policies from Phase 3          | Phase 3 output               |
| Verified simulation engine from Phase 4/5 | Phase 4-5 output             |
| <code>configs/simulation.yaml</code>      | Experiment configuration     |
| <code>configs/cbd_scenario.yaml</code>    | CBD experiment configuration |

**Output Files / Deliverables**

| File                                          | Description                       |
|-----------------------------------------------|-----------------------------------|
| results/tables/descriptive_statistics.csv     | Full descriptive statistics       |
| results/tables/anova_results.csv              | ANOVA results                     |
| results/tables/posthoc_comparisons.csv        | Tukey HSD post-hoc tests          |
| results/tables/confidence_intervals.csv       | 95% confidence intervals          |
| results/tables/effect_sizes.csv               | Cohen's d effect sizes            |
| results/tables/exp1_summary.csv               | Experiment 1 summary              |
| results/tables/exp2_pivot_rt.csv              | Experiment 2 pivot table          |
| results/tables/exp3_pivot_rt.csv              | Experiment 3 pivot table          |
| results/tables/exp4_pivot_rt.csv              | Experiment 4 pivot table          |
| results/tables/sensitivity_summary.csv        | Overall sensitivity summary       |
| results/tables/cbd_comparison.csv             | CBD comparison table              |
| results/tables/cbd_summary_all.csv            | CBD comprehensive summary         |
| results/tables/queue_statistics.csv           | Queue statistics                  |
| results/tables/queue_anova.csv                | Queue ANOVA results               |
| results/tables/seasonal_analysis.csv          | Seasonal variation analysis       |
| results/tables/table1-4_*.csv and *.tex       | Publication tables (CSV + LaTeX)  |
| results/figures/exp1-4_*.png                  | Experiment result figures         |
| results/figures/pub_fig1-5_*.png              | Publication-quality figures       |
| results/figures/queue_*.png                   | Queue analysis figures            |
| results/figures/seasonal_*.png                | Seasonal analysis figures         |
| results/figures/project_summary_dashboard.png | Summary dashboard                 |
| docs/experimental_design.md                   | Experimental design specification |
| docs/output_analysis.md                       | Output analysis documentation     |
| docs/cbd_robustness_analysis.md               | CBD robustness analysis report    |

| File                       | Description           |
|----------------------------|-----------------------|
| docs/queue_analysis.md     | Queue analysis report |
| docs/gap_closure_report.md | Gap closure report    |

Scripts / Notebooks Used

| Script / Notebook                       | Purpose                                        |
|-----------------------------------------|------------------------------------------------|
| scripts/run_production_experiments.py   | Execute 1,440 production runs (SEED_BASE=42)   |
| scripts/run_cbd_experiment.py           | Execute 330 CBD robustness runs (SEED_BASE=42) |
| scripts/analyze_production_results.py   | Statistical analysis of production results     |
| scripts/analyze_queue_metrics.py        | Queue metrics analysis                         |
| scripts/analyze_seasonal_patterns.py    | Seasonal variation analysis                    |
| scripts/generate_publication_figures.py | Generate publication-quality figures           |
| scripts/generate_summary_dashboard.py   | Generate summary dashboard                     |
| notebooks/07_production_results.ipynb   | Interactive production results exploration     |
| notebooks/08_statistical_analysis.ipynb | Interactive statistical analysis               |
| notebooks/09_cbd_analysis.ipynb         | CBD robustness analysis notebook               |

Dependencies

- **Depends on:** Phase 5 (verified/validated simulator), all previous phases

8. Phase 7 — Reporting & Final Delivery

Objective

Compile all findings into a comprehensive technical report, executive presentation, and complete project archive.



## **Step-by-Step Procedure**

| Step | Action                                                                                       | Output                |
|------|----------------------------------------------------------------------------------------------|-----------------------|
| 7.1  | Write comprehensive technical report covering all sections (Introduction through Appendices) | Technical report      |
| 7.2  | Add formal abstract (~250 words) and executive summary                                       | Report front matter   |
| 7.3  | Create List of Figures (44 figures) and List of Tables (24 tables)                           | Report indices        |
| 7.4  | Add “Why DES?” justification section                                                         | Methodology rationale |
| 7.5  | Embed inline conceptual model flowchart                                                      | Visual documentation  |
| 7.6  | Add reproducibility section (seeds, environment, instructions)                               | Reproducibility guide |
| 7.7  | Write executive presentation with key findings and recommendations                           | Presentation document |
| 7.8  | Prepare implementation roadmap (0-3, 3-6, 6-12, 12+ months)                                  | Roadmap               |
| 7.9  | Create file inventory and project archive documentation                                      | Archive manifest      |
| 7.10 | Perform Phase 21 compliance assessment                                                       | Compliance report     |
| 7.11 | Address any remaining gaps identified in assessment                                          | Gap remediation       |
| 7.12 | Generate PDF versions of all documentation                                                   | PDF deliverables      |
| 7.13 | Create Work Breakdown Structure documentation                                                | This document         |
| 7.14 |                                                                                              | Release               |

| Step | Action                           | Output |
|------|----------------------------------|--------|
|      | Final review and version tagging |        |

## Input Files

| File                              | Source          |
|-----------------------------------|-----------------|
| All results from Phases 1-6       | Previous phases |
| All documentation from Phases 1-6 | Previous phases |

## Output Files / Deliverables

| File                                       | Description                                                 |
|--------------------------------------------|-------------------------------------------------------------|
| docs/technical_report.md                   | Comprehensive technical report (v2.0.0, Phase 21 compliant) |
| docs/technical_report.pdf                  | PDF version of technical report                             |
| docs/executive_summary.md                  | Standalone executive summary                                |
| docs/executive_presentation.md             | Executive presentation                                      |
| docs/implementation_roadmap.md             | Implementation roadmap                                      |
| docs/final_summary.md                      | Final project summary                                       |
| docs/file_inventory.md                     | Complete file inventory                                     |
| docs/project_archive.md                    | Archive documentation                                       |
| docs/phase21_assessment.md                 | Phase 21 compliance assessment                              |
| docs/gap_remediation_plan.md               | Gap remediation plan                                        |
| docs/phase_comparison_summary.md           | Phase comparison summary                                    |
| docs/project_workflow_wbs.md               | This WBS document                                           |
| docs/cbd_comparison_and_validity_report.md | CBD comparison and validity report                          |
| docs/cbd_definition.md                     | CBD definition and boundary documentation                   |
| All PDF versions of above documents        | Generated PDFs                                              |
| README.md (updated)                        | Final project README                                        |

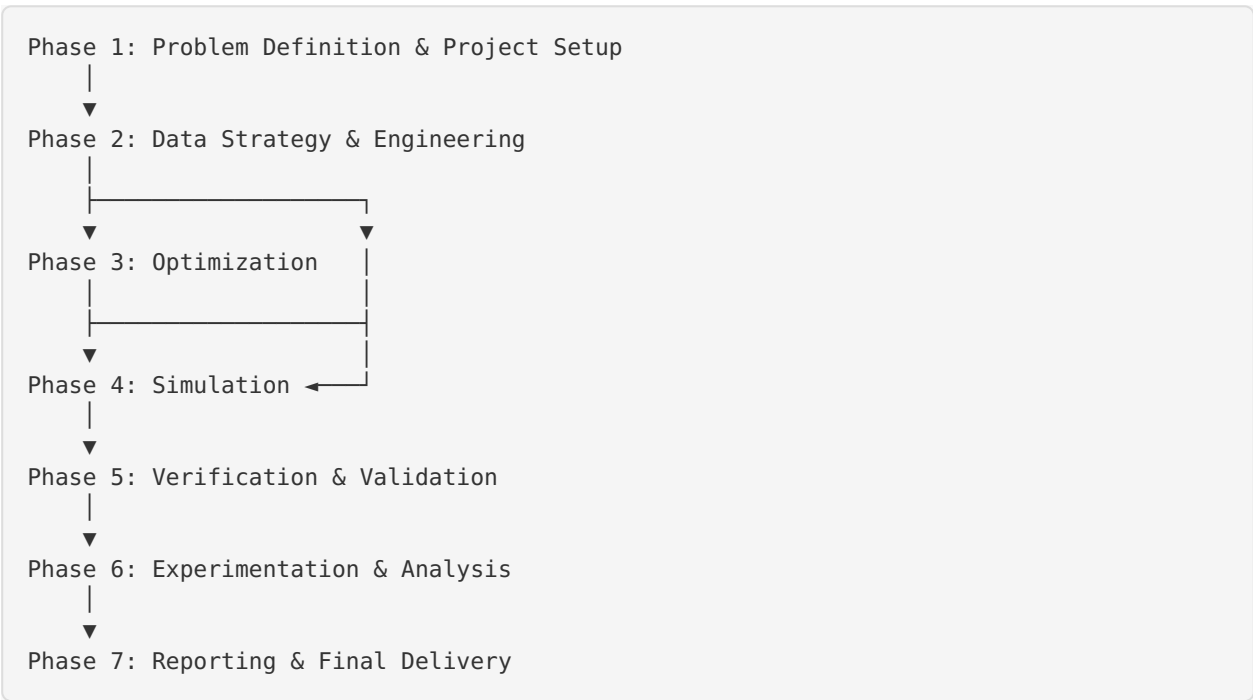
Scripts / Notebooks Used

| Script / Notebook              | Purpose                   |
|--------------------------------|---------------------------|
| Manual writing and compilation | Documentation authoring   |
| PDF generation pipeline        | Markdown → PDF conversion |

Dependencies

- **Depends on:** All previous phases (1–6)

9. Phase Dependency Diagram



**Critical Path:** Phase 1 → Phase 2 → Phase 4 → Phase 5 → Phase 6 → Phase 7

**Parallel Opportunities:**

- Phase 3 (Optimization) and Phase 4 (Simulation design) can partially overlap, as the conceptual model design does not depend on optimization results. However, the simulation implementation requires the allocation policies from Phase 3 as input.
- Documentation tasks in Phase 7 can begin concurrently with Phase 6 experimentation.

## 10. Timeline Summary

| Phase | Description                        | Estimated Duration | Cumulative |
|-------|------------------------------------|--------------------|------------|
| 1     | Problem Definition & Setup         | 1 week             | Week 1     |
| 2     | Data Strategy & Engineering        | 2 weeks            | Week 3     |
| 3     | Optimization & Policy Design       | 1 week             | Week 4     |
| 4     | Simulation Design & Implementation | 2 weeks            | Week 6     |
| 5     | Verification & Validation          | 1 week             | Week 7     |
| 6     | Experimentation & Analysis         | 2 weeks            | Week 9     |
| 7     | Reporting & Final Delivery         | 2 weeks            | Week 11    |

**Total Project Duration:** ~11 weeks

### Key Milestones

| Milestone                 | Target  | Deliverable                                             |
|---------------------------|---------|---------------------------------------------------------|
| M1: Data Ready            | Week 3  | Processed datasets, distance matrix, lambda tables      |
| M2: Policies Generated    | Week 4  | P0, P1, P2 allocation vectors for all K values          |
| M3: Simulator Operational | Week 6  | Working DES engine with all components                  |
| M4: V&V Complete          | Week 7  | Verified and validated simulation model                 |
| M5: Experiments Complete  | Week 9  | 1,770 production runs with statistical analysis         |
| M6: Final Submission      | Week 11 | Complete report, presentation, and reproducible archive |

## Appendix: Complete Script Inventory

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| Script                                     | Phase | Purpose                     | Key Parameters             |
|--------------------------------------------|-------|-----------------------------|----------------------------|
| scripts/<br>data_audit.py                  | 2     | Raw data quality audit      | —                          |
| scripts/<br>audit_step1_boundaries.py      | 2     | Boundary file audit         | —                          |
| scripts/<br>audit_step2_firehouses.py      | 2     | Firehouse data audit        | —                          |
| scripts/<br>audit_step3_precincts.py       | 2     | Precinct data audit         | —                          |
| scripts/<br>audit_step4_crashes.py         | 2     | Crash records audit         | —                          |
| scripts/de-<br>mand_modeling.py            | 2     | NHPP demand calibration     | base_rate=3.48             |
| scripts/<br>run_optimization_comparison.py | 3     | Run all optimization models | $K \in \{20, 30, 40, 48\}$ |
| scripts/<br>run_verification.py            | 5     | 4 verification tests        | seed=42                    |
| scripts/<br>run_validation_pilots.py       | 5     | 3 validation pilots         | seed_base=100, 200, 300    |
| scripts/<br>run_production_experiments.py  | 6     | 1,440 production runs       | SEED_BASE=42, R=30         |
| scripts/<br>run_cbd_experiment.py          | 6     | 330 CBD robustness runs     | SEED_BASE=42, R=30         |
| scripts/ana-<br>lyze_production_results.py | 6     | Statistical analysis        | —                          |
|                                            | 6     | Queue metrics analysis      | —                          |

| Script                                  | Phase | Purpose                     | Key Parameters |
|-----------------------------------------|-------|-----------------------------|----------------|
| scripts/analyze_queue_metrics.py        |       |                             |                |
| scripts/analyze_seasonal_patterns.py    | 6     | Seasonal variation analysis | —              |
| scripts/generate_publication_figures.py | 6     | Publication figures         | —              |
| scripts/generate_summary_dashboard.py   | 6     | Summary dashboard           | —              |

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